

design

design.md — The Lenny Growth Assistant

1. UI/UX Principles

1. **Trust through transparency** — the user should always be able to see what model answered and what it was grounded in. Nothing is hidden behind a generic “AI response.”
2. **Chat-first, artifact-second** — conversation is the primary surface; artifacts are a natural extension that appears alongside it, never a separate destination.
3. **Honest failure** — errors are stated plainly in the UI (“Local model isn’t running” beats a spinner that never resolves).
4. **No dead ends** — every state (loading, empty, error) tells the user what’s happening and, where possible, what to do next.

2. Information Architecture

```
App
├── Sidebar
│   ├── New Chat button
│   └── Session list (title = first user message, timestamp)
├── Main panel
│   ├── Header: provider indicator (Claude / Ollama), health dot
│   ├── Message thread
│   │   ├── User messages
│   │   └── Assistant messages (+ inline source citations, + "generate artifact" affordance)
│   └── Composer (input + send)
└── Artifact panel (appears beside/below main panel when an artifact exists)
    ├── Tab: Markdown | HTML
    └── Rendered artifact (sandboxed) + "download" / "copy" actions
```

3. Main Screens

- **Chat (default):** sidebar + message thread + composer. Artifact panel collapsed until first artifact is generated.
- **Chat with artifact open:** two-pane layout (chat left ~55%, artifact right ~45% on desktop); stacked on mobile (artifact below chat, expandable).

4. Chat Experience

- User sends a message → optimistic UI shows it immediately → assistant message streams in (or appears with a loading indicator if streaming isn’t used) → sources render as small citation chips beneath the assistant message, each naming the episode.
- Follow-up questions visually continue the same thread; no re-explaining context needed from the user.

5. Session Handling

- “New Chat” always starts a fresh, isolated session (own DB row, own retrieval context memory within that conversation).
- Sidebar lists prior sessions so users can resume; switching sessions swaps the visible thread without losing state in the other.

6. Source Citations

- Each grounded assistant message shows a compact row of citation chips (e.g., “Ep. 142 — Growth Loops”), clickable to expand the specific retrieved excerpt and, where available, a link to the source transcript file.
- If no chips appear, the message text itself states the knowledge base didn’t support a grounded answer — the *absence* of chips is intentional, not a bug state.

7. Loading States

- Composer disabled + subtle “thinking” indicator while a response is in flight.
- Artifact panel shows a skeleton/placeholder while generation is in progress, distinct from the chat loading indicator.

8. Error States

- Provider unavailable: inline banner in the message thread (“Local model (Ollama) isn’t reachable — start Ollama or switch providers”), not a raw error dump.
- Database unavailable: header shows a degraded-mode notice; chat still functions (best-effort, non-persistent) per architecture.md §10.
- Artifact generation failure: artifact panel shows a retry affordance, chat thread is unaffected.

9. Empty States

- New session with no messages: composer + a short set of example prompts (“Ask about onboarding activation,” “Ask about pricing experiments”) to orient a first-time evaluator.
- No artifacts yet: artifact panel shows a brief explainer of what it does rather than staying blank.

10. Artifact Viewer

- Renders **inside** the app (never redirects), matching the assignment’s explicit requirement.
- Markdown artifacts render through a sandboxed Markdown renderer (HTML pass-through disabled).
- HTML artifacts render inside a sandboxed `<iframe srcdoc>` (see architecture.md §8) with a visible “this preview is sandboxed — scripts are disabled” note, so the evaluator understands the security posture without reading the docs.
- Copy / download actions let the user take the artifact out of the app.

11. Responsive Behavior

- Breakpoint-based: desktop shows chat + artifact side-by-side; tablet/mobile stacks them vertically with the artifact collapsible to save space.
- Composer and header remain fixed/sticky at small widths.

12. Accessibility

- All interactive elements (send button, new chat, session items, artifact tabs) are keyboard-reachable and have visible focus states.
- Semantic HTML (`<button>`, `<nav>`, `<main>`) over generic `<div onClick>` patterns.
- Color is never the sole indicator of state (e.g., provider health uses both color and text label, not just a colored dot).
- Sufficient contrast ratios for text on the chosen color palette (verified against WCAG AA during implementation).

13. Keyboard Navigation

- `Enter` sends a message, `Shift+Enter` inserts a newline in the composer.
- `Tab` order: sidebar → header → message thread → composer → artifact panel.
- `Esc` collapses an expanded citation or closes the artifact panel on mobile.

14. Visual Hierarchy

- Assistant messages visually distinct from user messages (alignment + subtle background), but both readable at a glance.
- Citations are secondary in visual weight (smaller, muted) so they inform without competing with the answer itself.
- Provider indicator is persistently visible but unobtrusive — a small badge in the header, not a banner.

15. Design Decisions (rationale)

Decision	Rationale
Artifact panel beside chat, not a modal/new tab	Assignment explicitly requires in-app rendering, “not... redirecting to another application”
Citations shown by default, not behind a toggle	Grounding transparency is a core product requirement, not a nice-to-have
Explicit “sandboxed” note on HTML artifacts	Evaluator should understand the security posture without needing to read architecture.md separately
Degraded-mode chat when DB is down	Prioritizes availability of the core experience over strict consistency, appropriate for a demo/evaluation context — documented as a trade-off in the PRD